15 Lecture 15

In this lecture, we present one of the most important results of Probability Theory, the strong law of large numbers. In this lecture we use the notation $S_n = X_1 + \ldots + X_n$.

15.1 Law of large numbers

Let's start with a simple example. Assume that we repeatedly toss a fair coin. Our intuition is telling us that half of the tosses will be H and the other half will be T. To formulate this mathematically, consider the random variables $X_i = \begin{cases} 1, & \text{if the } i\text{-th toss is H,} \\ 0, & \text{otherwise.} \end{cases}$ We expect that

$$\frac{X_1 + \ldots + X_n}{n} \approx \frac{1}{2}, \quad n \in \mathbb{N}.$$

This is indeed what happens in the long run ($n \to \infty$) and it is justified by the law of large numbers.

Theorem 15.1 (Weak Law of Large Numbers)**.** *Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of independent and identically distributed random variables with finite variance ($\mathbb{E}[X_1^2] < \infty$). Then,*

$$\frac{X_1 + \ldots + X_n}{n} \xrightarrow{\mathbb{P}} \mathbb{E}[X_1], \quad \text{as } n \to \infty.$$

Proof. Let $\varepsilon > 0$. We set $\mu = \mathbb{E}[X_1]$. We need to show that $\mathbb{P}(|S_n/n - \mu| > \varepsilon) \xrightarrow{n\to\infty} 0$. We have by Chebyshev's inequality and the independence

$$\begin{aligned}
\mathbb{P}(|S_n/n - \mu| > \varepsilon) \leq \frac{\mathrm{Var}(S_n/n)}{\varepsilon^2} &= \frac{1}{\varepsilon^2}\mathbb{E}\left[\left(\frac{(X_1-\mu)+\ldots+(X_n-\mu)}{n}\right)^2\right] \\
&= \frac{1}{\varepsilon^2 n^2}\left(\sum_{i=1}^n \mathbb{E}[(X_i-\mu)^2] + \sum_{i\neq j}\mathbb{E}[(X_i-\mu)(X_j-\mu)]\right) \\
&= \frac{1}{\varepsilon^2 n^2}\left(n\mathrm{Var}(X_1) + \sum_{i\neq j}\mathbb{E}[X_i-\mu]\cdot\mathbb{E}[X_j-\mu]\right) \\
&= \frac{\mathrm{Var}(X_1)}{\varepsilon^2 n} \xrightarrow{n\to\infty} 0.
\end{aligned}$$

The proof is complete. □

We can improve the type of convergence to almost sure convergence.

Theorem 15.2 (Strong Law of Large Numbers)**.** *Under the assumptions of Theorem 15.1, we have*

$$\frac{X_1 + \ldots + X_n}{n} \xrightarrow{a.s.} \mathbb{E}[X_1], \quad \text{as } n \to \infty.$$

Remark 15.3. The above result holds even if the assumption $\mathbb{E}[X_1^2] < \infty$ (finite variance) is substituted by the more general $\mathbb{E}[|X_1|] < \infty$ (recall that $L^2 \subset L^1$). We will prove it only under the assumption $\mathbb{E}[X_1^2] < \infty$.

Proof. Set $\mu = \mathbb{E}[X_1]$ and $\mathrm{Var}(X_1) = \sigma^2$. We will prove the result in the case where the random variables X_n take values in $[0, +\infty]$. We also set $Y_n = \frac{S_n}{n} - \mu$. Our goal is to show that $Y_n \xrightarrow{\text{a.s.}} 0$.

As we computed in Theorem 15.1,

$$\mathbb{E}[Y_n] = 0 \quad \text{and} \quad \mathbb{E}[Y_n^2] = \frac{\sigma^2}{n}.$$

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This implies

$$\mathbb{E}\left[\sum_{n=1}^{+\infty} Y_{n^2}^2\right] = \sum_{n=1}^{+\infty} \mathbb{E}[Y_{n^2}^2] = \sum_{n=1}^{+\infty} \frac{\sigma^2}{n^2} < \infty,$$

hence $\sum_{n=1}^{+\infty} Y_{n^2}^2 < \infty$ almost surely, consequently

$$\lim_{n\to\infty} Y_{n^2} = 0, \text{ almost surely.}$$

We will now derive $\lim_{n\to\infty} Y_n = 0$. Let $k \geq 1$ and $n(k) = \lfloor \sqrt{k} \rfloor$. Then,

$$\frac{S_{n(k)^2}}{(n(k)+1)^2} \leq \frac{S_k}{k} \leq \frac{S_{(n(k)+1)^2}}{n(k)^2}.$$

However,

$$\frac{S_{n(k)^2}}{(n(k)+1)^2} = \frac{S_{n(k)^2}}{n(k)^2}\left(\frac{n(k)}{n(k)+1}\right)^2 = Y_{n(k)^2}\left(\frac{n(k)}{n(k)+1}\right)^2 \xrightarrow{k\to\infty} \mu$$

and

$$\frac{S_{(n(k)+1)^2}}{n(k)^2} = Y_{(n(k)+1)^2}\left(\frac{n(k)+1}{n(k)}\right)^2 \xrightarrow{k\to\infty} \mu.$$

Therefore $S_k/k \xrightarrow{\text{a.s.}} \mu$ as $k \to \infty$.

In the more general case where X_n can take negative values as well, we write

$$X_n = \max\{X_n, 0\} - \max\{-X_n, 0\} =: X_n^+ - X_n^-.$$

We notice that $X_n^+, X_n^- \in [0, +\infty]$, hence

$$\frac{X_1 + \ldots + X_n}{n} = \frac{X_1^+ + \ldots + X_n^+}{n} - \frac{X_1^- + \ldots + X_n^-}{n} \xrightarrow{\text{a.s.}} \mathbb{E}[X_1^+] - \mathbb{E}[X_1^-] = \mathbb{E}[X_1].$$

□

Remark 15.4. An even more general statement is that the strong law of large numbers holds even if $\mathbb{E}[X_1] = +\infty$ or $-\infty$. We omit the proof in our lectures.

Example 1 (Coin toss)
Assume that $X_i \sim \text{Ber}(1/2)$, as in the introductory example. The law of large numbers implies

$$\frac{X_1 + \ldots + X_n}{n} \xrightarrow{\text{a.s., } n\to\infty} \mathbb{E}[X_1] = \frac{1}{2}.$$

Example 2 (Monte Carlo methods)
A Monte Carlo method is any algorithm that exploits the law of large numbers to compute a (deterministic) quantity. This is best motivated by an example.

Suppose that we want to approximate the value of π. We recall that π is the area of a unit circle. That is, letting $C = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$,

$$\pi = \int_{-1}^{1}\int_{-1}^{1} \mathbf{1}_C(x, y)\, dxdy.$$

In other words, if $X, Y \sim U([-1, 1])$ and are independent,

$$\mathbb{E}[\mathbf{1}_C(X, Y)] = \int_{-1}^{1}\int_{-1}^{1} \mathbf{1}_C(x, y) f_{X,Y}(x, y) dxdy = \frac{1}{4}\int_{-1}^{1}\int_{-1}^{1} \mathbf{1}_C(x, y) dxdy = \frac{\pi}{4}.$$

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If we consider a sequence of independent pairs (X_n, Y_n) such that $X_n, Y_n \sim U([-1,1])$, then the law of large numbers gives

$$\lim_{n\to\infty} \frac{\mathbf{1}_C(X_1,Y_1) + \ldots + \mathbf{1}_C(X_n,Y_n)}{n} = \frac{\pi}{4}, \quad \text{almost surely.}$$

This gives us an algorithm to approximate π:

1. Pick $n \in \mathbb{N}$ and set S_n to zero.
2. Generate two independent random variables X, Y from the uniform distribution $U([-1,1])$. If $X^2 + Y^2 \leq 1$, increment S_n by one.
3. Repeat step 2 n times. Then, $\frac{\pi}{4} \approx \frac{S_n}{n}$.

A Python code for this procedure is given below:

```python
import random

def estimate_pi(n):
    S_n = 0  # This step 1

    for _ in range(n):
        # Here is step 2 (and 3)
        X = random.uniform(-1, 1)
        Y = random.uniform(-1, 1)

        # Check if the point is inside the unit circle
        if X**2 + Y**2 <= 1:
            S_n += 1

    pi_estimate = 4 * S_n / n
    return pi_estimate

# Example usage
n = 1000000
pi_approx = estimate_pi(n)
print(pi_approx)
```

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An important question is how big n has to be in order to obtain a good approximation, that is $|4S_n/n - \pi| \leq \varepsilon$ for some given ε. We can estimate n such that this inequality happens with very high probability using Chebyshev's inequality. In particular,

$$\mathbb{P}(|4S_n/n - \pi| > \varepsilon) \leq \frac{\mathrm{Var}(4S_n/n)}{\varepsilon^2} = \frac{4^2}{n^2\varepsilon^2}\mathrm{Var}(\sum_{i=1}^n \mathbf{1}_C(X_i,Y_i)) = \frac{16}{n\varepsilon^2}\mathrm{Var}(\mathbf{1}_C(X_1,Y_1)) \approx \frac{1}{n\varepsilon^2}.$$

Example 3 (Bernstein polynomials)
Assume that $(X_n)_{n\in\mathbb{N}}$ are i.i.d with $X_i \sim \mathrm{Ber}(p)$, and $f\colon [0,1] \to \mathbb{R}$ is a continuous function. Using $(X_n)_{n\in\mathbb{N}}$ we will construct a sequence $(p_n)_{n\in\mathbb{N}}$ of polynomials such that $p_n \to f$ uniformly. This is a known fact from real analysis and is called *Stone-Weirestrass* theorem.

Let $\varepsilon > 0$. Since f is continuous and $[0,1]$ is compact, it is also uniformly continuous. Therefore, there exists $\delta > 0$ such that $|f(x) - f(y)| < \varepsilon$ for any x, y such that $|x - y| < \delta$. Also, for any $p \in [0,1]$ we have

$$\mathbb{E}\left[\left|f\left(\frac{S_n}{n}\right) - f(p)\right|\right] \leq \varepsilon\mathbb{P}\left(\left|\frac{S_n}{n} - p\right| < \delta\right) + 2\|f\|_\infty \mathbb{P}\left(\left|\frac{S_n}{n} - p\right| \geq \delta\right) \leq \varepsilon + 2\|f\|_\infty \frac{p(1-p)}{n\delta^2},$$

where in the last inequality we used Chebyshev's inequality and the calculation we did in the proof of Theorem 15.1. Since $p(1-p) \leq 1/4$, we deduce

$$|\mathbb{E}[f(S_n/n)] - f(p)| \leq \mathbb{E}\left[\left|f\left(\frac{S_n}{n}\right) - f(p)\right|\right] \leq \varepsilon + 2\|f\|_\infty \frac{1}{4n\delta^2}.$$

Since the bound is independent of p, we get $\mathbb{E}[f(S_n/n)] \to f(p)$ uniformly in p. Finally, we observe that $S_n = X_1 + \ldots + X_n \sim \mathrm{Bin}(n, p)$, hence

$$\mathbb{E}[f(S_n/n)] = \sum_{k=0}^{n} f\left(\frac{k}{n}\right)\binom{n}{k} p^n (1-p)^{n-k}.$$

These expressions are polynomials in p, they approximate (uniformly) f and are called *Bernstein polynomials*.

Example 4 (Hypothesis testing)
Imagine a new candy vending machine. Each time you press the button, it dispenses a red candy with probability p, and a candy of another color with probability $1-p$. You suspect that the machine does not give red candies 50% of the time, and you want to check this claim experimentally.

Let $X_i = \begin{cases} 1 & \text{if the } i\text{-th candy is red,} \\ 0 & \text{otherwise.} \end{cases}$, for $i \in \mathbb{N}$. We perform the experiment n times and we define the *sample mean*

$$\hat{p}_n = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

This is our estimate for p. As in the previous example, Chebyshev's inequality gives a bound on the probability that our estimate deviates from the true p:

$$\mathbb{P}(|\hat{p}_n - p| \geq \varepsilon) \leq \frac{\mathrm{Var}(\hat{p}_n)}{\varepsilon^2} \leq \frac{1}{4n\varepsilon^2}.$$

Suppose we want to be within $\varepsilon = 0.05$ of the true p with probability at least 0.95:

$$\frac{1}{4n(0.05)^2} \leq 0.05 \implies n \geq 2000.$$

Thus, sampling roughly 2000 candies provides a reasonable guarantee that $\hat{p}_n$ is close to p.

Remark 15.5. By using a stronger estimate for $\mathbb{P}(|\hat{p}_n - p| \geq \varepsilon)$ we might be able to achieve this level of certainty with less trials. For example, since $X_i \in [0, 1]$, Hoeffding's inequality (you can have a look at Problem 7.2 from our file) gives:

$$\mathbb{P}(|\hat{p}_n - p| \geq \varepsilon) \leq 2\exp(-2n\varepsilon^2),$$

which allows us to achieve the same confidence with fewer samples. Indeed, if we want $|\hat{p}_n - p| < 0.05$ with probability at least 0.95, set the right-hand side to be less or equal to 0.05:

$$2\exp(-2n(0.05)^2) \leq 0.05 \Leftrightarrow \exp(-0.005n) \leq 0.025 \Leftrightarrow -0.005n \leq \log(0.025) \implies n \geq \frac{-\log(0.025)}{0.005} \approx 738$$

Thus, using Hoeffding's inequality, we only need approximately 738 samples to achieve the same confidence, compared to 2000 samples with Chebyshev.

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